Approach and future work

I built a TypeScript document-to-entry review workflow: users upload shipment PDFs, confirm extraction, inspect evidence beside filing fields, resolve open items and export deterministic NetCHB XML. The supported profile is ocean entry type 01. Other profiles stay blocked instead of appearing complete. [VERIFIED: server/src/checks, web/src]

The backend uses Gemini for JSON observations, with a value, evidence snippet/page, status and confidence for every field. File hashes, model settings and prompt/schema identity key the disk cache. Schema v3 adds routing, containerization, consignee, manufacturer and assist-linkage observations; older reviews remain readable. Corrections and accepted suggestions are stored separately from the original response. [VERIFIED: shared/src/schemas, server/src/extraction]

I traced NetCHB's entry vocabulary through the located entry.xsd and data_type.xsd, preserving originals and validating generated XML with xmllint-wasm. Source discovery and access limitations are recorded in docs/research.md. The model never generates XML. Integer-cent calculations and explicit decisions feed a deterministic serializer. Schema validity is separate from local readiness; neither guarantees external acceptance. [VERIFIED: spec/, server/src/builder; UNKNOWN: live NetCHB acceptance]

Generalization comes from entity-based schemas and decisions, rather than sample-specific rules. Versioned port/country/unit tables offer reviewable mappings; HTS search offers candidates, not automatic classification. The sample retains invoice arithmetic, packing quantity, origin and weight discrepancies. Audit caught OCR errors in a factory tax number, master B/L and handwritten container; source-backed corrections preserve the original output. Missing broker identifiers and classifications remain explicit in the draft and its open-items report. [VERIFIED: shared/src/suggestions.ts, server/src/reference, submission/]

AI helped with research, implementation and tests; model-produced ground truth was treated as an unverified transcription. Regression tests cover synthetic non-apparel shipments, conflicts, stale decisions, cache isolation, API and UI contracts. One sample cannot establish broad extraction accuracy. The complete AI chat history accompanies the submission separately. [VERIFIED: server/test, docs/ai-log.md; UNKNOWN: unseen-document accuracy]

With more time, I would collect independently labelled shipments across commodities and measure field/table accuracy; add other entry/transport profiles and commodity-specific agency/trade-remedy checks; validate HTS unit inheritance and MID construction; and refresh references by effective date. These improve coverage and reduce manual research. Authenticated encrypted storage, retention controls, broader browser automation and a NetCHB test-account integration would address operational reliability. SPI/state expansion and a licensed SCAC registry would verify applicability and identity beyond format checks. [INFERRED: improvements based on current limits]